

Building A 2-Meter Antenna

The Roll Up J Pole

ABSTRACT

Building a 2-meter J-pole antenna from twin lead ladder line is a simple and inexpensive project that yields a portable and effective antenna. A J-pole is a half-wave vertical antenna with an integrated quarter-wave matching stub which eliminates the need for radials or a ground plane. Built from flexible materials, this antenna is well suited for portable use.

This guide specifies solid core, 450-Ohm ladder line, a heavy-duty variation, that is durable and easy to work with. Supplies are currently limited.

MATERIALS

The design is very simple – ladder line and a section of coax with an antenna connector.

Start with approximately 60 inches of 450-ohm ladder line and a convenient length of 50-ohm coax, such as RG-58, RG-8X, or RG-174. Cut one end off of a pre-built cable or attach a connector, such as PL-259 or BNC, depending on your radio.

DESIGN

The wavelength in meters of the frequency of our favorite repeater, 146.94 MHz, is the speed of light divided by the frequency, $300/146.94$, giving 2 meters. One quarter of 2 meters is 50 cm, or about 20 inches.

The J-pole antenna is an end-fed half-wave antenna that is matched to the feed line by a shorted quarter-wave stub. A half-wave antenna fed at one end has a very high input impedance. A shorted quarter-wave stub has a similar high impedance at its open end, making a good match to the antenna. Any value of input impedance can be obtained by connecting the feed line to the proper point along the matching stub.

One arm of the pair is extended a half wavelength to make the antenna. By attaching the antenna's feed line to the other arm at the proper point, the stub will transform this impedance down to match the lower feed line impedance.

CONSTRUCTION

Cut a piece of 450-ohm ladder line to a length of approximately 60 inches.

Strip about 1/2 to 3/4 inches of insulation from both wires at one end of the ladder line. Twist and solder the exposed copper wires together to form a solid, shorted connection.

Measure up 19 inches from the shorted end. Cut and remove one of the two conductors and the attached web of plastic insulation. Only the single, unbroken wire should continue to the top. This creates the characteristic "J" shape of the antenna.



Measure up 2 inches from the shorted bottom end. Carefully remove a small section of insulation on both wires, about 1/8 to 1/4 inches, to expose the copper underneath. This is where your coax cable will connect.



Prepare your feed line by stripping the outer insulation to expose the shield braid and inner conductor. Keep the exposed sections as short as possible.

Solder the coax center conductor to the long element of the J-pole. Solder the coax shield braid to the short element of the J-pole. Use electrical tape or heat-shrink tubing to insulate and weather-seal the connection point.

Use a cable tie to secure the coaxial cable to the ladder line just below the feed point. This prevents the weight of the coax from stressing your soldered connections.



To prevent common mode current from traveling down the outside of the coax shield, create a simple choke. Wrap about 4 turns of coax in a 5 inch coil. Secure it with cable ties or tape.

At the very top of the antenna, fold the last inch of the long element into a loop and secure it with a cable tie. This provides a convenient way to hang the antenna for use.

MOUNT THE ANTENNA

Rather than using a mast, this antenna is suitable for hanging from a line or a hook.

TUNE THE ANTENNA

While the dimensions provided are close, the final tuning depends on your specific materials and environment. An antenna analyzer, such as the NanoVNA, is the best way to view SWR and impedance.

Hang the antenna by the loop. Keep the antenna away from metal objects and as high as possible.

Connect your antenna analyzer to the coax and measure the SWR across the 2-meter band. Note the frequency where the SWR is lowest.

If the lowest SWR frequency is too low, the antenna is too long. Trim the top of the long element in very small increments, 1/8 inch at a time, and re-check the SWR. If the lowest SWR frequency is too high, the antenna is too short. To fix this, you may need to start over or add length. Adjust the feed point.

If the impedance is higher than desired, you can also slide the coax connection point slightly up or down the matching stub. Move it in small increments, 1/8 inch at a time, until you find 50 Ohms.